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“Time to market is a critical factor for design engineers”

The new oscilloscopes are evolving in multiple areas, but the most interesting aspect seems to be the increased focus on the user—the engineer who actually uses the scope.

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What are the most exciting changes affecting scopes recently?

The oscilloscopes market is characterised by three distinct categories—basic, main and professional. In the mainstream segment, users are mostly embedded engineers working on a mixed-domain design. In the analogue-to-digital era, a lot of digital designs came onto the analogue board, thus bringing up the Mixed Signal Oscilloscopes (MSOs) several years ago. The trend has evolved since, and what we are seeing today is that this mixed domain has moved to another level with the inclusion of wireless RF signals. Analogue, digital and now RF signals are coming together to the embedded domain, creating significant changes on how work is done in designs houses and research centres, specifically with respect to analysing those signals in as short a time as possible.

How is this affecting engineers' design work?

RF signals are also now being added to the mix of digital and analogue signals that have to be analysed. This is not easy for the design engineers working on the project, because they have to put all of these together and analyse on multiple fronts to successfully debug and complete a design. Time to market is a critical factor for design engineers, since the product lifecycle period is significantly shorter today. It means they have to come out with a newer version of their design three months down the line, which also results in a shorter innovation cycle. Thus the engineers need tools where time to answer is shorter for them.

What are the elements that such a tool should have?

With integrated instruments, engineers get multiple tools in one instrument in a compact form factor, and are thus able to get results faster. The key aspect of the integration in an MDO3000 is the inclusion of a logic analyser, protocol analyser, spectrum analyser, function generator and digital voltmeter into the oscilloscope itself. This is cru-

cial from the customers' perspective. Their tool of choice to debug and verify their designs is the oscilloscope. The other instruments are used on an intermittent basis.

So is it relevant only to those working on debugging on multiple domains?

Multiple instruments are not only about the ability to debug in multiple domains, it's also about a design engineer who has the time to debug thoroughly. With a mixed-domain oscilloscope, he is able to use two methodologies to debug—the frequency domain approach or the time domain approach. Previously, he could use only time domain. With two options to debug or validate his design, the chances of his design passing through or his bringing the product into market, is much higher. Instruments like these are relevant to a broader group of design engineers, both education and research people. The multiuse features of MDO3000, for instance, offer the functionality of six independent instruments without the significant cost of separate instruments.

How exactly is this instrument beneficial for a design team starting out?

The basic customer will continue to use the basic scope as even design houses still have a combination of basic, mid-stream and performance scopes. When you look at the ROI aspect, there is an expectation that the upgradability built into the instrument should make it scalable for the future. In fact, the upgradability of instruments like this new mixed-domain oscilloscope sets it apart in the marketplace due to both the number of upgrades available as well as how easy the upgrade process is. There are two major categories of upgrades. First, there are performance upgrades. Customers can upgrade the bandwidth of their oscilloscope all the way up to 1GHz. When the analogue bandwidth is upgraded, the frequency range of the included spectrum analyser is also upgraded to the new bandwidth level.